

Independent FCI and FCI Plus from Source Fidelity to Tennessee STAR

**Paper-aligned algorithms. Known-truth validation.
A cautious empirical application.**

An auditable evidence chain—not one definitive causal DAG.

ALGORITHM EVIDENCE

Original sources → Independent code → Known truth

APPLICATION EVIDENCE

STAR experiment → Arm contrasts → PAG diagnostics

EVIDENCE HIERARCHY

Randomized contrast

Stable adjacency

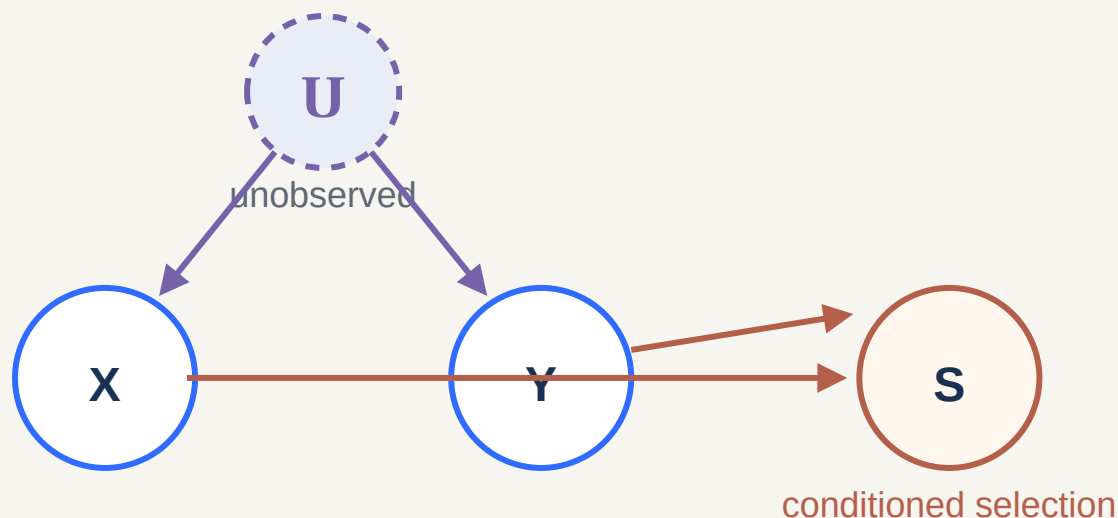
Unresolved marks

Interpret downward
with increasing caution

Latent Variables and Selection Change the Target

PROBLEM FRAME

UNDERLYING CAUSAL
SYSTEM



Marginalizing U and conditioning on S can induce a mixed ancestral relation among observed variables.

RIGHT OUTPUT

TARGET

PAG, not one unique DAG

○ → circle → arrow

⊥ → tail → arrow

↔ bidirected

Adjacency $\neq$ direct effect

Circle = unresolved endpoint

Sample CI test $\neq$ exact oracle

**The graph records invariant ancestral constraints—
and preserves uncertainty where the data cannot decide.**

The Project Asks Five Auditable Questions

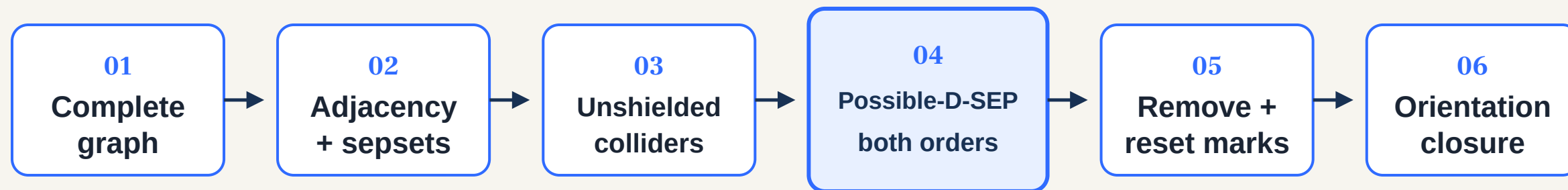
- 1** **THEORY**
What do the two original sources actually assume and prove?
- 2** **TRACEABILITY**
Where is each paper stage implemented and tested?
- 3** **KNOWN TRUTH**
What happens when the correct graph is known?
- 4** **SOFTWARE**
Which comparisons across implementations are fair?
- 5** **APPLICATION**
Which STAR findings are randomized, stable, or exploratory?

Source fidelity → validation → application interpretation.
An attractive real-data graph cannot reverse this order.

Original FCI Closes Non-Local Separations

SPIRTEs ET AL. 2000

HISTORICAL SIX-STAGE SCHEDULE



Local adjacency search can miss a separating set that lies outside current neighborhoods.

POSSIBLE-D-SEP PATH CONDITION

Every intermediate triple is either
a collider on the current path, or part of a triangle.

This computable superset makes non-local separator search possible.

THEOREM 6.4 — ORACLE SCOPE

Faithfulness + correct CI answers

Sound partially oriented
inducing-path graph.

Maximal informativeness not proved there.

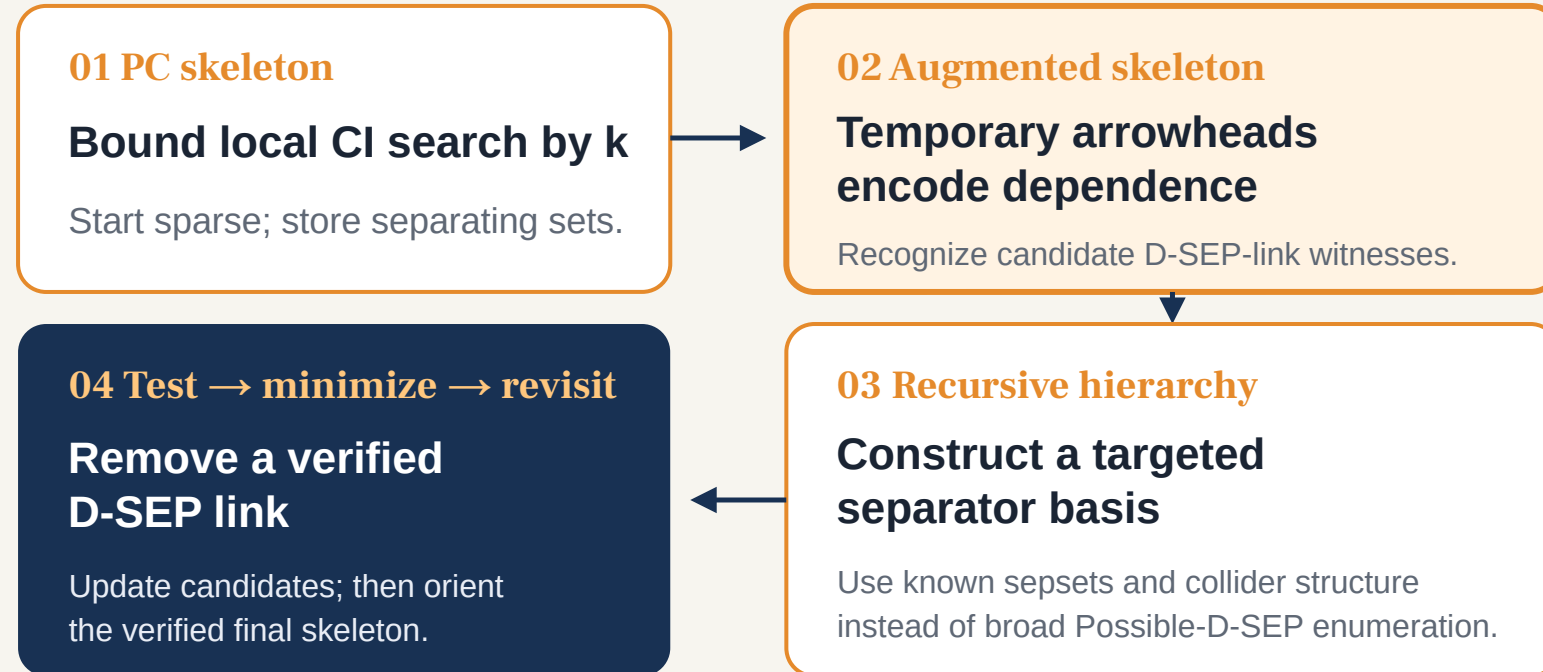
Spirtes, Glymour & Scheines (2000), §6.7, printed pp. 144–145 · Possible-D-SEP p. 144 · Theorem 6.4 p. 145

Modern PAG orientation uses Zhang's later rule set.

FCI Plus Uses a Sparse Logical Characterization

CLAASSEN ET AL. 2013

ALGORITHM 2: FROM LOCAL SKELETON TO D-SEP LINK REMOVAL



CONDITIONAL QUERY BOUND

$$O(N^{2k+4})$$

Only under:

- fixed observed-MAG degree k
- Faithfulness
- constant-time exact CI oracle

NOT a sample-accuracy guarantee
NOT a wall-clock guarantee

D-SEP WITNESS

$$U \leftrightarrow X \leftrightarrow Y \leftrightarrow V$$

+ hierarchy-derived separating information

Every Paper Stage Is Inspectable and Testable

TRACEABILITY

SOURCE	IMPLEMENTATION BOUNDARY	TEST / AUDIT EVIDENCE
Spirtes FCI Adjacency + sepsets Possible-D-SEP orientation closure	skeleton.py · pdsep.py orientation.py · rules.py FCI.fit · possible_dsep refine_skeleton_with_pdsep	Exact-oracle PAG recovery Possible-D-SEP unit paths historical schedule regression
Claassen FCI Plus Augmented skeleton D-SEP links + hierarchy bounded endpoint bases	dsep.py · fci_plus.py build_augmented_skeleton possible_dsep_links · hierarchy refine_skeleton_with_fci_plus_dsep	Figure 4(b) separator fixture candidate-link witness tests hierarchy and query-bound checks
Audit layer Engineering extension not a source theorem	PAG · FCIResult · exports CI/orientation trace · sepset source edge explanations · order/bootstrap audit	PAG invariants + provenance serialization regressions cyclic-order and stability artifacts

Reusable API `fci(data, ...)` · `fci_plus(data, ...)` · `FCIPlusConfig.paper` / `practical`

Paper and Robust Profiles Stay Separate

PAPER FCI PLUS PROFILE

Source schedule

- Explicit degree bound k
- First separating set
- Non-stable skeleton schedule
- Zhang-rule orientation schedule

`FCIPlusConfig.paper(k=...)`

VS

ROBUST APPLICATION PROFILE

Finite-sample policy

- Stable skeleton + max-p sepsets
- Conservative collider decisions
- Restricted tail orientation
- Order and bootstrap diagnostics

`FCIPlusConfig.practical(...)`

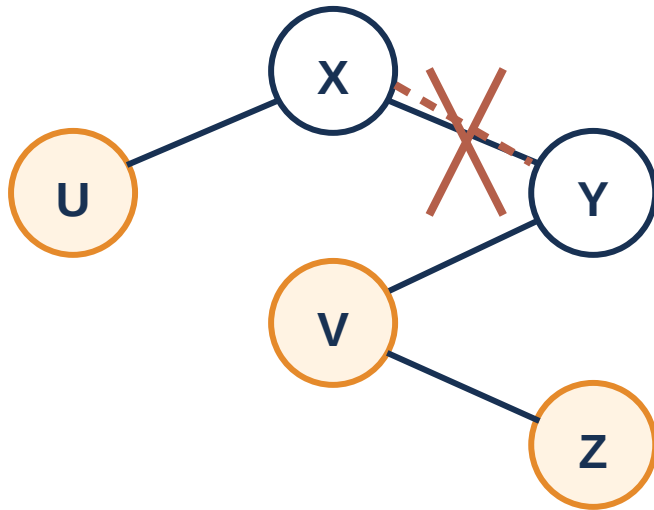
PREDECLARED TRADE-OFF

More CI work + more circle endpoints in exchange for **stable skeletons + conservative marks**

Known Truth Separates Logic from Sample Error

CLAASSEN FIGURE 4(b) FIXTURE

A non-local separator removes $X \rightarrow Y$



Exhibited separator: {U, V, Z}

TWO LEVELS OF KNOWN-TRUTH EVIDENCE

Exact oracle · Figure 4(b) logical CI queries



Fixed- α regression · $\alpha = .001$ for both · 5 families $\times$ 3 seeds

Endpoint-sensitive exact-edge F1



Skeleton F1: .979 for both profiles

Mean CI calls: 197.1 $\rightarrow$ 380.7 · N = 2,500

Fewer oracle queries $\neq$ finite-sample superiority.

The source prints a MAG;
the repository derives the PAG target independently.

An Auditability Comparison, Not a Leaderboard

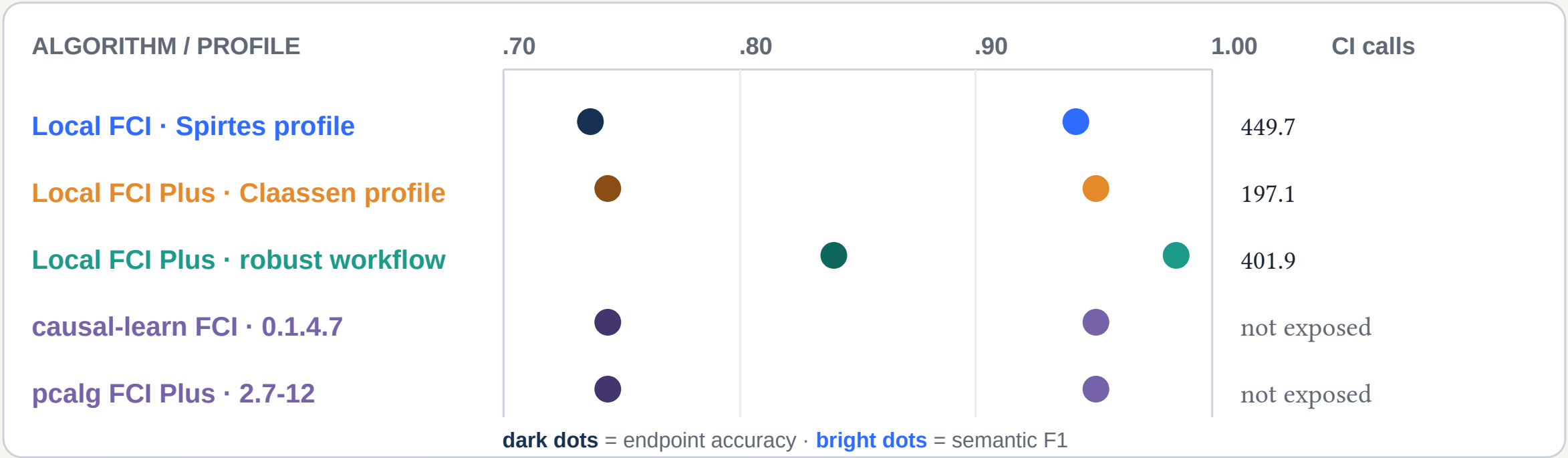
TOOL / EVIDENCE	FCI	FCI Plus	CI TRACE	ORDER AUDIT	BOOTSTRAP	EXPORT	STATUS
fci_engine 0.1.0	YES	YES	structured	built in	built in	artifacts	RUN
pcalg 2.7-12	YES	YES	not established	partial	not established	NO	RUN
causal-learn 0.1.4.7	YES	NO	not exported	not established	not established	NO	RUN
Tetrad 7.6.10	YES	NO	logger only	partial	workflow	NO	DOCS ONLY

LOCAL DISTINCTION
Explicit paper profiles + CI/orientation traces + separator provenance
+ stability audits + reproducible artifact exports

Matched Benchmark: Trade-offs, Not Dominance

KNOWN-TRUTH SOFTWARE BENCHMARK

5 families × 3 repeats N = 2,500 75 / 75 runs $\alpha = .001$ paper/reference · .01 robust auto



ROBUST WORKFLOW
semantic F1 .985 · endpoint accuracy .840

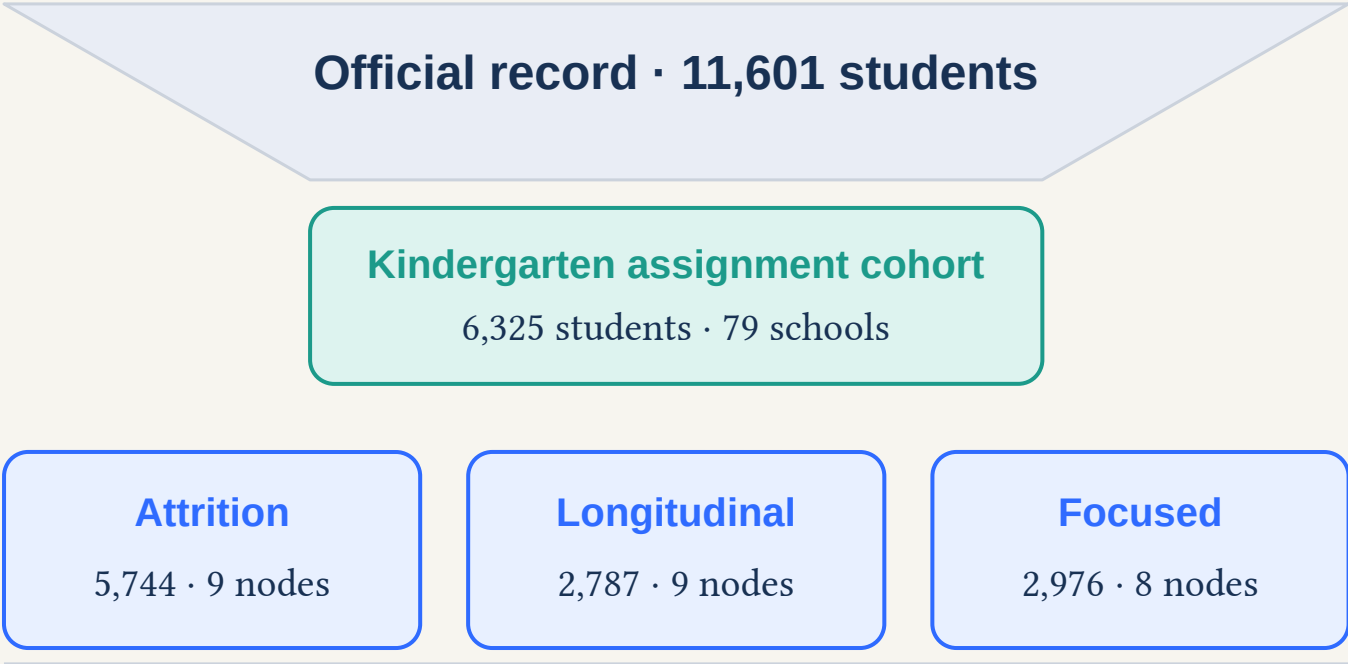
INTERPRETATION LIMIT
Different α policy: workflow evidence,
not isolated algorithm superiority.

STAR Supplies Randomization—and Selection Risk

APPLICATION BOUNDARY

DESIGN-BASED EXPLORATORY PAG

COHORT CONSTRUCTION



IDENTIFICATION BOUNDARY

Experiment first

Kindergarten arm assignment supplies the strongest design basis.

PAG second

Later complete-case panels may encode attrition and conditioning effects.

No temporal constraints supplied to the discovery algorithms

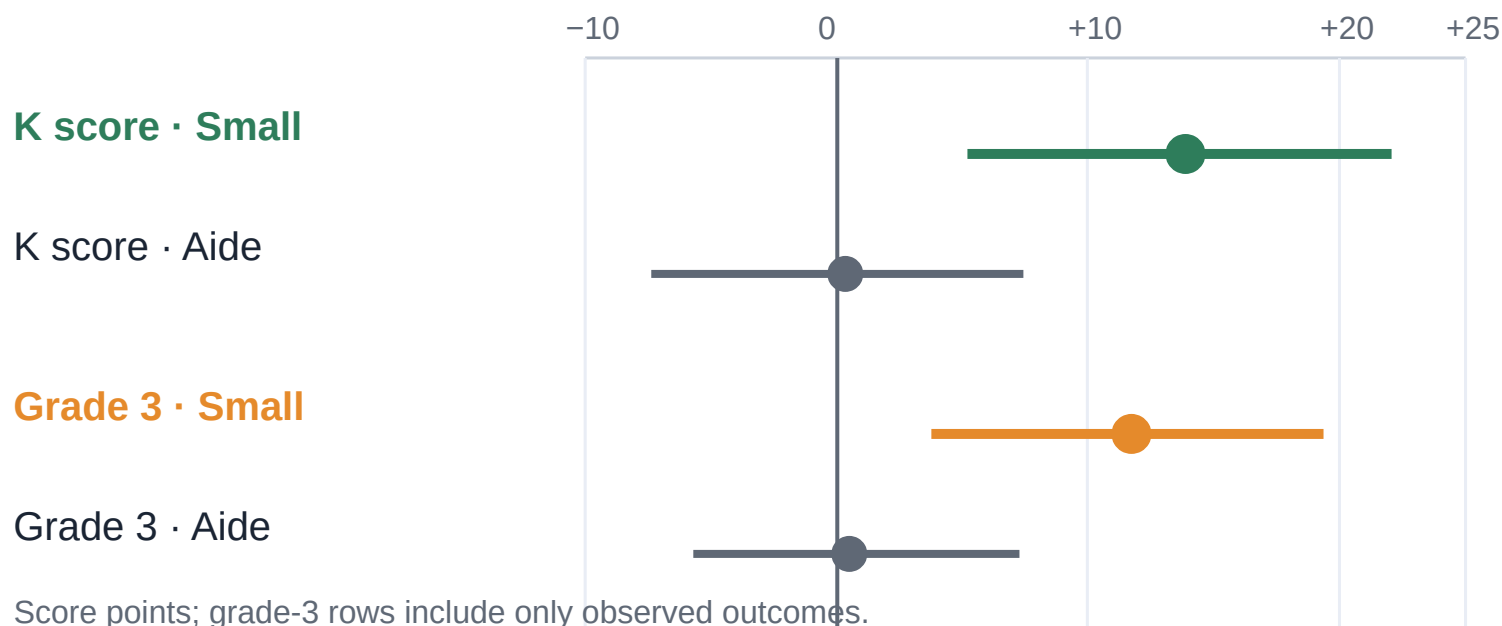
1,000 school-cluster resamples for arm contrasts · **12 school resamples** for coarse PAG stability only

The Strongest Signal Comes from Randomization

STAR ARM CONTRASTS

■ RANDOMIZED K CONTRAST ■ OBSERVED GRADE-3 FOLLOW-UP

ARM MINUS REGULAR · SCHOOL-CLUSTER BOOTSTRAP 95% INTERVAL



DESIGN-BASED READOUT

13.90 points

Small – regular, kindergarten
95% interval [5.21, 22.07]

0.31 points

Aide – regular, kindergarten
95% interval [-7.40, 7.38]

Observed-arm contrast supports
a possible small-class benefit

Observed grade-3 score, Small – Regular: **+11.69 [3.74, 19.38]** — descriptive, not long-run ITT.

Grade-3 observation rate: Small – Regular **+1.54 pp [-1.73, 4.56]** · Aide – Regular **-2.05 pp [-5.35, 1.41]**

6,325 kindergarten-assignment cohort · 1,000 school-cluster resamples · star_descriptive_contrasts.csv

Stable Adjacencies Outrank Endpoint Directions

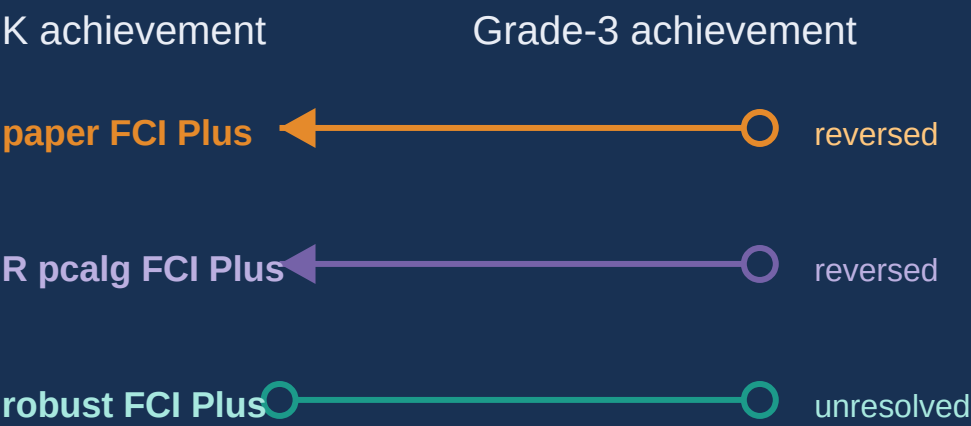
■ STABLE STRUCTURE ■ DIRECTION CONFLICT ○ ROBUST ABSTENTION

ORDER-SHIFT REPRODUCTIONS

tested shifts	FCI	paper FCI Plus	robust
Attrition	2 / 9	9 / 9	9 / 9
Longitudinal	8 / 9	5 / 9	9 / 9
Focused	6 / 8	6 / 8	8 / 8

Robust total: 26 / 26 · temporal flags: 0

LONGITUDINAL ENDPOINT COMPARISON

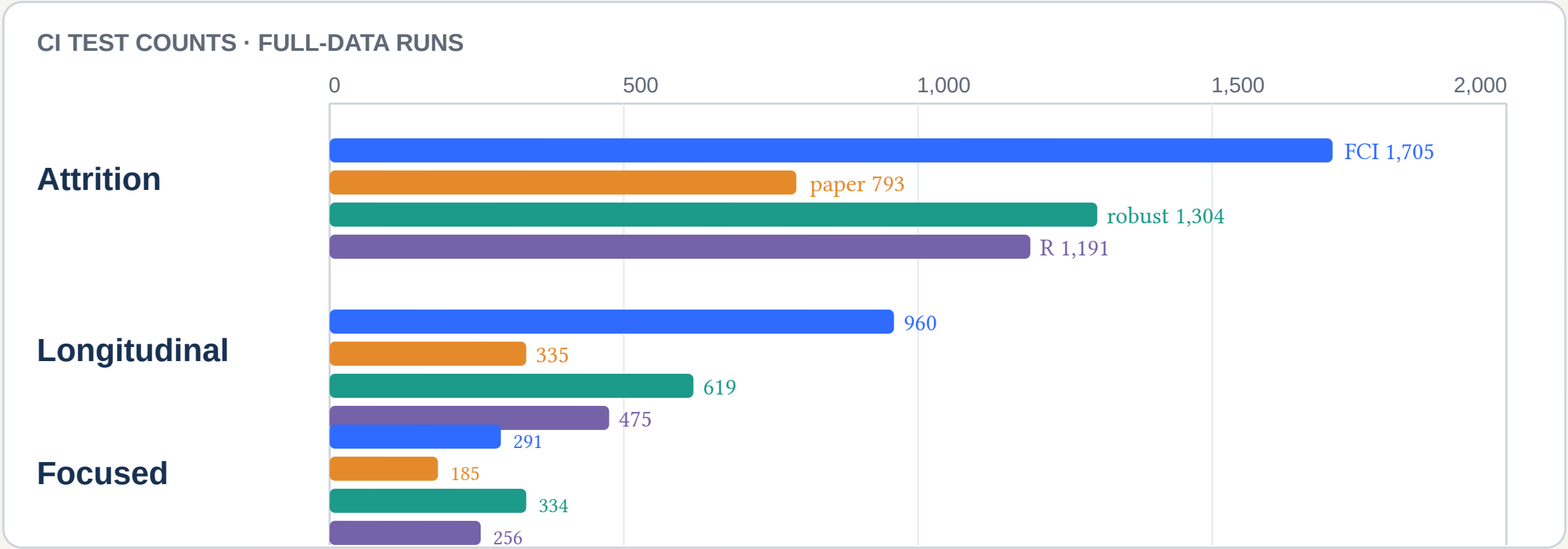


A circle endpoint is calibrated abstention—
not additional causal identification.

STABLE ADJACENCIES

Early–later achievement and early achievement–observation: 100% local bootstrap frequency.

FCI Plus Reduces CI Work, Not Causal Uncertainty



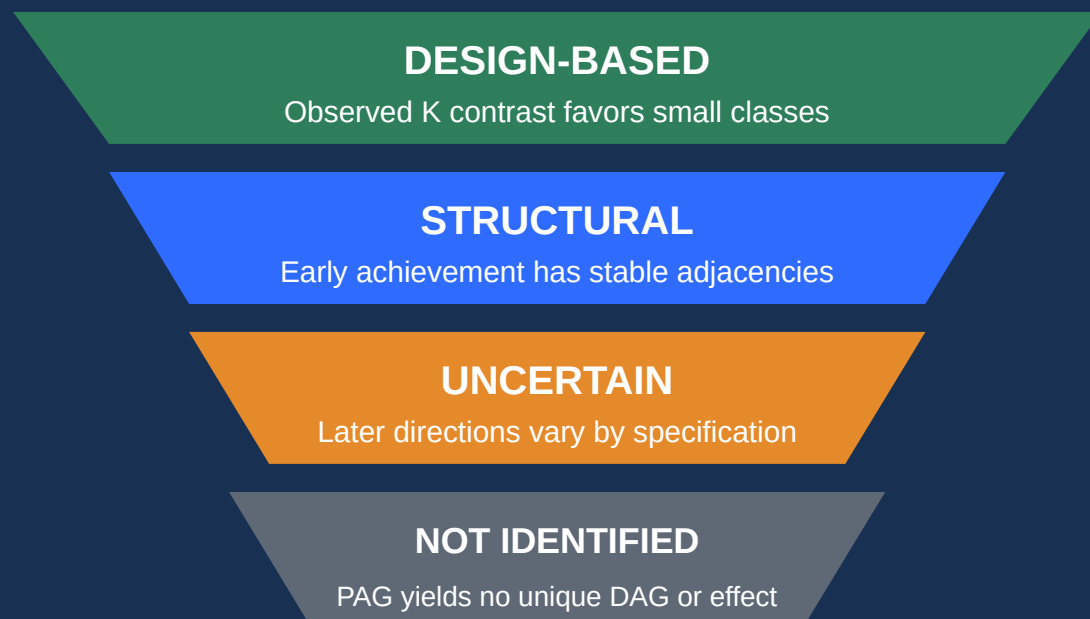
Paper FCI Plus / FCI CI work: 47% attrition · 35% longitudinal · 64% focused

Median seconds [FCI / paper / robust / R] — A: 5.44/1.79/2.84/10.39 · L: 2.15/.34/.56/1.65 · F: .32/.16/.25/.88

From Source Text to Cautious Evidence

Randomized evidence remains primary; the PAG is a disciplined uncertainty audit.

DEFENSIBLE STAR EVIDENCE HIERARCHY



COMPLETED DELIVERABLES

Paper profiles + source map

FCI and FCI Plus stages traced to code and tests

Known-truth + software audits

Exact oracle, simulation, pcalg, causal-learn

STAR report + reproducible artifacts

Data, figures, LaTeX report, deck, and notes

NEXT VALIDATION More CI regimes → larger benchmarks → formal randomized analysis → external replication

Discussion